

The Visual System

The hierarchy of cortical visual processing

Computational Neuroscience

University of Bristol

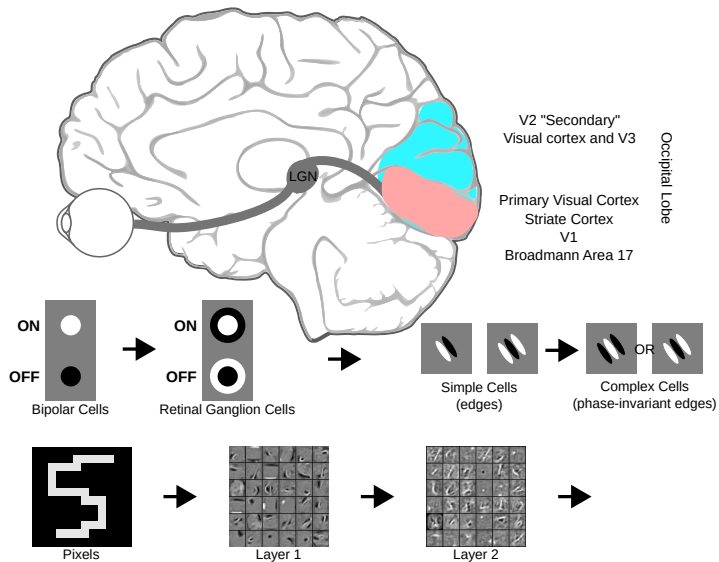
mrule

Learning outcomes:

- ▶ Describe the transformation of visual input from the retina to V1 and onward into more abstract cortical areas
- ▶ Describe how receptive fields and tuning properties change with increasing depth into the cortical visual processing hierarchy
- ▶ Briefly describe the distinction between “what” and “where” pathways in visual processing

The hierarchy of cortical visual processing

Previous slides reviewed the transformation of visual input
from retinal on/off cells through V1 complex cells



Now...

“What” vs “Where”

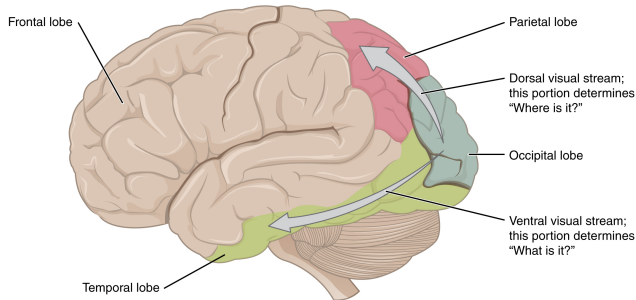


illustration by OpenStax College

Two-streams hypothesis Primates split visual processing into 2 specialised pathways

ventral stream, ‘what’ object identity, semantics

- ▶ temporal lobe, language & memory

dorsal stream, ‘where’ location, geometry, speed

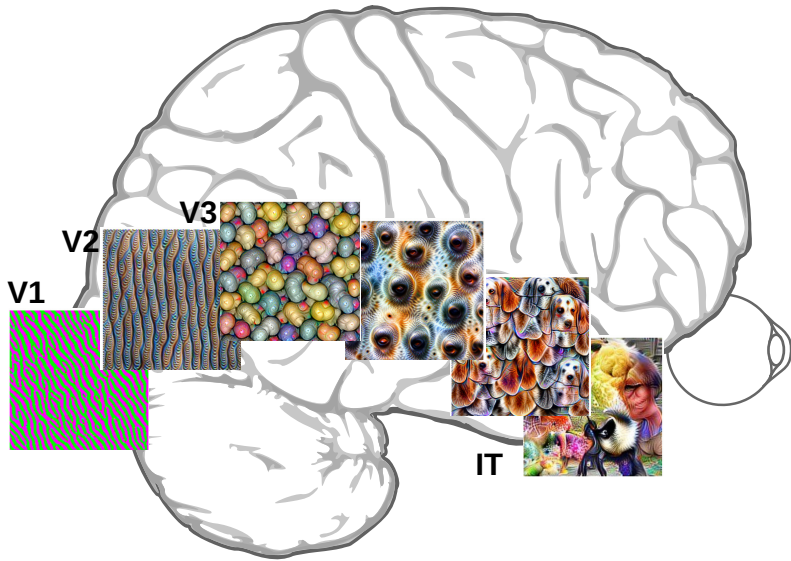
- ▶ parietal lobe ↔ motor cortex

Coding along the visual hierarchy

Receptive fields change across visual-processing hierarchy

With increasing depth, receptive fields become:

- ▶ Larger
 - Pool together larger regions of visual input
 - More invariant to the specific location of an input
- ▶ Sensitive to more complicated aspects of the visual stimulus
 - e.g. recognising specific objects, or certain complicated motions
- ▶ More multi-modal (i.e. also depend on non-visual sensory signals)
 - Multi-sensory integration in parietal cortex
- ▶ More sensitive to top-down, contextual information
 - Task context, animal's behavioural state, attention, etc.



neural network receptive fields OpenAI microscope via Götz-Hahn et al. (2023)

The hierarchy of cortical visual processing

Neurons sensitive to an object's movement, irrespective of its identity, are more likely to be found in the _____ or “where” cortical processing stream.

The _____ ventral (“what”) processing stream is responsible for recognising, naming, and remembering visually encountered objects based on their identity.

Receptive fields become (more/less) _____ spatially invariant with increasing depth into the cortical visual processing hierarchy

Would you expect V1 to be (more/less) _____ amenable to top-down modulation compared to area IT, deep in the ventral processing stream?

A cortical representation of an object's position, shape, and relative location to the hand, would be better decoded from (parietal/temporal/occipital) _____ lobe.

The hierarchy of cortical visual processing

Neurons sensitive to an object's movement, irrespective of its identity, are more likely to be found in the dorsal or “where” cortical processing stream.

The temporal lobe ventral (“what”) processing stream is responsible for recognising, naming, and remembering visually encountered objects based on their identity.

Receptive fields become (more/less) more spatially invariant with increasing depth into the cortical visual processing hierarchy

Would you expect V1 to be (more/less) more amenable to top-down modulation compared to area IT, deep in the ventral processing stream?

A cortical representation of an object's position, shape, and relative location to the hand, would be better decoded from (parietal/temporal/occipital) parietal lobe.

end

